Fact: A perfect $\mathbb{F}_p$-alg. ($x \mapsto x^p$ iso)

(1) $\exists!$ ring $W(A)$ which is p-adically complete $\bar{w}$ $W(A)/pW(A) \cong A$. Def. "ring of Witt vectors". $W(\mathbb{F}_p) = \mathbb{Z}_p$.

($W(A) \cong \varprojlim W(A)/p^n W(A)$)

(2) if B p-adically complete ring, $\bar{f}: A \to B/pB \rightsquigarrow \exists!\ \hat{f}: A \to B$, $f: W(A) \to B$ st.

$$\begin{tikzcd} W(A) \arrow[r] \arrow[d, "f"'] & A \cong W(A)/pW(A) \arrow[d, "\bar{f}"] \arrow[dl, dashed, "\exists!"'] \\ B \arrow[r] & B/pB \end{tikzcd}$$

ex. $B = W(A)$. $g: A \xrightarrow{\sim} W(A)/pW(A) \rightsquigarrow \hat{g}: A \to W(A)$

Def. $a \in A \rightsquigarrow$ "Teichmüller lift" $[a] := \hat{g}(a) \in W(A)$.

ex. • $r \in \mathbb{Z}_{>0}$. $W(\mathbb{F}_{p^r}) \cong$ valuation ring of deg r unramified extn of $\mathbb{Q}_p$.

• $W(\overline{\mathbb{F}}_p) \cong$ valuation ring of $\widehat{\mathbb{Q}_p^{un}}$, $\mathbb{Q}_p^{un}$ max. unramified extn of $\mathbb{Q}_p$.

Prop. A - perfect $\mathbb{F}_p$-alg.

(1) $\alpha \in W(A) \overset{\exists!}{\rightsquigarrow} a_0 \in A$ st. $\alpha \in [a_0] + pW(A)$

(2) $\alpha \in W(A) \overset{\exists!}{\rightsquigarrow} \alpha = \sum_{n=0}^{\infty} [a_n] p^n$, $a_n \in A$. "Teichmüller expansion" $\not\Leftrightarrow$ not the same as p-adic expansion for $A = \mathbb{F}_p$.

(3) $\varphi_A: A \to A$, $x \mapsto x^p$ $\overset{\exists!}{\rightsquigarrow}$ $\varphi_{W(A)}: W(A) \to W(A)$, $\sum_{n=0}^{\infty} [a_n] p^n \mapsto \sum_{n=0}^{\infty} [a_n^p] p^n$ "Frobenius of $W(A)$"

Proof. (1) $W(A) \to W(A)/pW(A) \cong A$, $\alpha \mapsto [a_0] \leftrightarrow a$

(2) inductively: given $\alpha \in \sum_{i=0}^{n} [a_i] p^i + p^{n+1} W(A)$, say $= \cdots + p^{n+1} \alpha_{n+1}$,

apply 1 to $\alpha_{n+1} \in W(A) \overset{\exists!}{\rightsquigarrow} a_{n+1} \in A$ st. $\alpha_{n+1} \in [a_{n+1}] + pW(A)$

so $p^{n+1}\alpha_{n+1} \in [a_{n+1}] p^{n+1} + p^{n+2} W(A)$

and $\alpha \in \sum_{i=0}^{n+1} [a_i] p^i + p^{n+2} W(A)$.

(3) $\varphi_A: A \xrightarrow{\sim} A \cong W(A)/pW(A) \xrightarrow[\text{fact (2)}]{\exists!} \varphi_{W(A)}: W(A) \xrightarrow{\sim} W(A)$. formula holds by construction. $\square$

Witt vectors/A may be constructed as sequences $(x_0, x_1, x_2 \ldots)$ of elts in A.

• Witt polynomials
$w_n = \sum_{i=0}^{n} p^i X_i^{p^{n-i}} = X_0^{p^n} + pX_1^{p^{n-1}} + p^2 X_2^{p^{n-2}} + \cdots$

• Thm. $\Phi \in \mathbb{Z}[X,Y] \rightsquigarrow \exists!\ (\phi_0 \ldots) \in \mathbb{Z}[X_0 \ldots Y_0 \ldots]$ st. $w_n(\phi_0 \ldots) = \Phi(w_n(X_0 \ldots), w_n(Y_0 \ldots))$

Prop. A pft $\mathbb{F}_p$-alg. $\alpha = \sum_{n=0}^{\infty} [a_n] p^n$, $\beta = \sum_{n=0}^{\infty} [b_n] p^n \in W(A)$.

(1) $\alpha + \beta = (a_0 + b_0) + \left(a_1 + b_1 - \frac{(a_0^{\frac{1}{p}} + b_0^{\frac{1}{p}})^p - a_0 - b_0}{p}\right) p + \cdots$

(2) $\alpha\beta = a_0 b_0 + (a_0 b_1 + a_1 b_0) p + \cdots$ $\square$

$R = k$ perfect field of char p. ($x \mapsto x^p$ iso)

Def. $\sigma := \varphi_{W(k)}$.

(1) $D, D' \in W(k)$-mod. $r \in \mathbb{Z}$, $f: D \to D'$ is σ^r-semilin. if $f(cm) = \sigma^r(c) f(m)$.

(2) Dieudonné module is finite free $D \in W(k)$-mod $\bar{w}$ σ-semilin $\varphi_D: D \to D$ "Frobenius" st. $pD \subset \operatorname{im}(\varphi_D)$.

(3) D, D' Dieudonné. $f \in \operatorname{Hom}_{W(k)}(D, D')$ is Dieudonné morphism if $f \circ \varphi_D = \varphi_{D'} \circ f$.

Lem. $W(k)$ is complete DVR $\bar{w}$ res. field k, uniformizer p.

Proof. follows from 2 general facts + $W(k) \cong \varprojlim W(k)/p^n W(k)$ + $W(k)/pW(k) \cong k$.

• $\alpha \in W(k)$ unit $\Leftrightarrow [a_0] \in k$ unit

• $[a_0] \in k \neq 0 \Leftrightarrow \alpha \in pW(k)$.

$\alpha \in W(k) = p^n u$ for $n \geq 0$, $u \in W(k)^\times$. $\square$

Prop. D Dieudonné $W(k)$-mod.

(1) φ_D inj

(2) coker φ_D is k-v.s.

(3) $\exists!$ σ^{-1}-semilin. $\psi_D: D \to D$ st. $\varphi_D \circ \psi_D = \psi_D \circ \varphi_D = [p]_D$ $\square$ "Verschiebung"

(4) $D^\vee = \operatorname{Hom}_{W(k)}(D, W(k))$ is naturally Dieudonné + $\varphi_{D^\vee}(f)(m) = \sigma(f(\psi_D(m)))$ for $f \in D^\vee$, $m \in D$. $\square$

Thm. (!) (Dieudonné) additive equiv $\{p\text{-div. gps}/k\}^{op} \xrightarrow{\sim} \{\text{Dieudonné } W(k)\text{-mod}\}$ st.

$G \mapsto \mathbb{D}G$ "Dieudonné crystal of G"

$\text{"}\varinjlim G_v\text{"}$

(i) $\operatorname{rk} \mathbb{D}G = \operatorname{ht} G$

(ii) $\dim_k \operatorname{coker} \varphi_{\mathbb{D}G} = \dim G$

(iii) $[p]_G \longrightarrow [p]_{\mathbb{D}G}$

(iv) $\mathbb{D}(G^\vee) \cong (\mathbb{D}G)^\vee$.

Pf sketch. key fact: $\forall n \geq 1$, $\exists$ k-gp sch W_n st. $W_n(A) \cong W(A)/p^n W(A)$ $\forall$ pft k-alg A.

if $G^\vee$ cnctd, $\mathbb{D}G := \varprojlim_v \varinjlim_n \operatorname{Hom}_{k\text{-Grp}}(G_v, W_n)$ $\bar{w}$ Frob. induced by φ_G.

if $G^\vee$ ét, i.e. G cnctd, $\mathbb{D}G := D(G^\vee)^\vee$

in general, recall connected-étale sequence splits (k perfect field) so $G \cong G^{unip} \times G^{mult}$ $\longrightarrow$ $\mathbb{D}G = \mathbb{D}G^{unip} \oplus \mathbb{D}G^{mult}$. $\square$

($)^\vee$ cnctd, ($)^\vee$ ét.

ex. • $\mathbb{D}(\mathbb{Q}_p/\mathbb{Z}_p) \cong W(k)$, $\varphi = \sigma$

• $\mathbb{D}(\mu_{p^\infty}) \cong W(k)$, $\varphi = p\sigma$.

Def. $K_0(k) := W(k)[\frac{1}{p}]$.

(1) $\exists!$ "Frobenius" automor. extending σ.

(2) isocrystal $/K_0(k)$ is f.d. $K_0(k)$-v.s D w/ σ-semilin. aut φ_D "Frob. of D"

(3) D, D' isocrystals $\rightsquigarrow$ $f \in \mathrm{Hom}_{K_0(k)}(D, D')$ is morphism of isocrystals if $f \circ \varphi_D = \varphi_{D'} \circ f$.

ex. · D Dieudon $W(k)$-mod $\rightsquigarrow$ $D[\frac{1}{p}] = D \otimes_{W(k)} K_0(k)$ isocrystal $/K_0(k)$ with Frob. $\varphi_D \otimes 1$.

· D, D' $K_0(k)$-isocrys $\rightsquigarrow$ $D \otimes_{K_0(k)} D'$ $K_0(k)$-isocrystal w/ Frob. $\varphi_D \otimes \varphi_{D'}$.

· D $K_0(k)$-isocrys $\rightsquigarrow$ $D^\vee := \mathrm{Hom}_{K_0(k)}(D, K_0(k))$ $K_0(k)$-isocry w/ Frob. $\varphi_{D^\vee}(f)(v) = \sigma \circ f \circ \varphi_D^{-1}(v)$ $\quad f \in D^\vee, v \in D$.

· D " " rk r $\rightsquigarrow$ $\det D = \bigwedge^r D$ rk 1 $K_0(k)$-isoc. (fix iso $\det D \cong K_0(k)$ as $K_0(k)$-mod)

Def. $D \neq 0$ $K_0(k)$-isocry.

(1) $\deg(D) \in \mathbb{Z}$ is unique st. $\varphi_{\det D}(1) \in p^{\deg D} W(k)^\times$ actually independent

(2) slope of D $\mu(D) := \frac{\deg(D)}{\mathrm{rk}(D)}$.

ex. $r > 0$, $d \in \mathbb{Z}$ $\rightsquigarrow$ D_λ, $\lambda := \frac{d}{r}$ reduced, "simple $K_0(k)$-isocrys of slope λ" has basis $e_1, \dots, e_r$ and

$\varphi_{D_\lambda}(e_i) = e_{i+1}$ $\quad i < r$

$\varphi_{D_\lambda}(e_r) = p^d e_1$.

Prop. G p-div gp$/k$.

(1) rk $\mathbb{D}G[\frac{1}{p}]$ = ht G

(2) deg $\mathbb{D}G[\frac{1}{p}]$ = dim G

(3) natural iso $\mathbb{D}(G^\vee)[\frac{1}{p}] \xrightarrow{\sim} \mathbb{D}(G)[\frac{1}{p}]^\vee \otimes_{K_0(k)} D_1$. □

Thm (!) (Manin) every $K_0(\bar{k})$-isocrys D $\overset{\exists!}{\rightsquigarrow}$ $D \cong \bigoplus_{i=1}^n D_{\lambda_i}$ $\quad \lambda_i \in \mathbb{Q}$. □

Prop. G p-div gp $/\bar{k}$.

(1) for $\mathbb{D}G[\frac{1}{p}] \cong \bigoplus_{i=1}^n D_{\lambda_i}$, $\quad 0 \le \lambda_i \le 1$.

(2) G ét $\iff$ $\lambda_i = 0$ $\forall i$

(3) G cnctd $\iff$ $\lambda_i > 0$ $\forall i$.

Proof. (1) $D := \mathbb{D}G[\frac{1}{p}]$. ETS $D_{\frac{d}{r}} \hookrightarrow D \Rightarrow 0 \le d \le r$. ($\frac{d}{r}$ reduced)

· $e := e_1 \in D_{d/r} \subset D$ satisfies $\varphi_D^r(e) = p^d e$.

· by replacing e w/ some $p^m e$, $m \ge 0$, can assume $e \in \mathbb{D}G$, $p^{-1}e \notin \mathbb{D}G$.

· since $\varphi_D|_{\mathbb{D}G}$ is Frob. of $\mathbb{D}G$, $p^{-1}e \notin \mathbb{D}G \Rightarrow d \ge 0$.

· since $p\varphi_D^{-1}$ is Vers. of $\mathbb{D}G$, $p^{r-d}e = p^r \varphi_D^{-r}(e) \Rightarrow r - d \ge 0$.

(2) $\underbrace{\deg D}_{= \dim G} = \sum_{i=1}^n \deg D_{\lambda_i}$, $\quad \deg D_{\lambda_i} \ge 0$.

we know G ét $\iff$ $\dim G = 0$ (cnctd-ét. seq splits)

(3) $\Leftarrow$: $\lambda_i > 0 \Rightarrow G^{ét} = 0$ by (2) as $D \cong \mathbb{D}(G^{ét})[\frac{1}{p}] \oplus \mathbb{D}(G^\circ)[\frac{1}{p}]$.

$\Rightarrow$: contrapos. if $\exists \lambda_i = 0$, $D \supset D_0$ contains e st. $\varphi_D(e) = e$. wlog $e \in \mathbb{D}G$.

get $W(k) \to \mathbb{D}G$, $\alpha \mapsto \alpha \cdot e$ nonzero morphism of Dieudonné modules.

$\Updownarrow$

nonzero morphism of p-div gps $G \to \mathbb{Q}_p/\mathbb{Z}_p$ (étale), as $\mathbb{D}(\mathbb{Q}_p/\mathbb{Z}_p) \cong W(k)$.

$\Downarrow$

G not cnctd. □

ex. E elliptic curve$/\mathbb{F}_p$.

(1) E ordinary $\rightsquigarrow$ $\mathbb{D}(E[p^\infty])[\frac{1}{p}] \cong D_0 \oplus D_1$

as neither cnctd nor étale.

(2) E supersingular $\rightsquigarrow$ $\mathbb{D}(E[p^\infty])[\frac{1}{p}] \cong D_{\frac{1}{2}}$

cnctd, so

rmk (Honda, Tate) D $\mathbb{F}_p$-isocrys is $\cong \mathbb{D}(A[p^\infty])[\frac{1}{p}]$ for A abelian variety $/\mathbb{F}_p$

$\Updownarrow$

$D \cong \bigoplus_{i=1}^n D_{\lambda_i}$, $\quad 0 \le \lambda_i \le 1$, $\lambda_i + \lambda_{n-i} = 1$.